



BIT1123

OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 – INDIVIDUAL (20%)

FACULTY OF INFORMATION TECHNOLOGY

CITY UNIVERSITY MALAYSIA

(CYBERJAYA CAMPUS)

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TABLE OF CONTENT:

1. Introduction.....	03
2. Knowledge gained from each tutorial.....	03-04
3. Challenges encountered.....	04-05
4. How the challenges were overcome.....	05
5. Improvements in Java programming skills.....	06
6. Understanding of Object-Oriented Programming concepts.....	06-07
7. Future learning plans.....	07
8. Conclusion.....	07
9. GitHub Repository URL.....	07-08

1. Introduction

This report describes the reflection on my learning process in the entire set of Object-Oriented Programming (OOP) tutorials taken throughout the semester. During ten weeks of practice, I acquired valuable knowledge in Java programming and OOP principles, ranging from basics of language syntax to developing a full-fledged GUI application. This portfolio is meant to be a record of my learning process and skills.

2. Knowledge Acquired from Each Tutorial

Week 1: Java Basics

Acquired fundamental knowledge about Java syntax, variables, and data types

Understood control structures (if-else statements, loops)

Developed simple applications, such as HelloWorld and StudentGrade

Set up development environment using VS Code

Week 2: Classes and Objects

Acquired knowledge of creating classes and objects

Got familiar with instance variables and methods

Practiced in object-oriented approach

Developed Student class with attributes and methods

Week 3-4: Introduction to Inheritance

Learning inheritance through extends keyword

Use of super for accessing parent class members

Inheritance hierarchy for Person, Student, and Lecturer classes

Method overriding

Week 5: Student Information System

Applying inheritance in a practical system

Encapsulation using getters and setters

Complete student management system

Week 6: Employee and Lecturer Inheritance

Advanced inheritance

Constructor chaining

Polymorphism using Employee and Lecturer classes

Week 7: Smart Home System

Application of polymorphism

Differences between method overloading and overriding

Dynamic method dispatch

Smart Home Appliance Management System

Week 8-9: File Input/Output and Exception Handling

File input/output using Java

Try-catch statements

Custom exceptions

Task.txt file handling

Week 10: Quiz Battle Game

Creation of GUI using Java Swing

Event driven programming

Use of all OOP concepts

Questions.java and QuizBattleGUI.java

3. Challenges Encountered

Challenge 1: Polymorphism Concept

At first, I faced some challenges in comprehending polymorphism. I had issues knowing when to apply superclass references to subclass objects. The dynamic method dispatch part was difficult to grasp in the Smart Home project.

Challenge 2: GUI Programming

Creating the Quiz Battle Game was difficult since it involved working with several GUI elements such as buttons, labels, and panels at the same time.

Challenge 3: File Input/Output Handling

Handling exceptions and reading and writing to files required practice. Issues came up in regards to file paths and exceptions in week 8-9.

Challenge 4: Git and GitHub

The concept of version control using Git was confusing for me. Commands such as add, commit, and push were not easy for me at first.

4. How Were the Challenges Resolved

Resolving Polymorphism Challenge

To resolve the polymorphism challenge, I developed small programs where I could test out inheritance and analyze how methods were resolved during the process using a debugger.

Resolving GUI Challenge

For the GUI, I broke the process into different parts including laying out the GUI elements, adding event listeners, and implementing quiz functionalities.

Resolving File I/O Challenge

In order to resolve the file I/O challenge, I looked at examples on how to use file I/O, practiced it on sample files and learned to use absolute paths first and later relative paths. I also learned about Java's exception hierarchy.

Resolving Git Challenge

I practiced Git commands, watched tutorial videos and used the Git interface within VS Code to visualize what changes I was making.

5. Improvements in Java Programming Skills

Through the tutorials, I have made significant improvements in the following areas:

Code Structure: Have been able to structure my code properly with good naming and packaging with the help of our lecturer.

Problem-Solving: Developed a systematic process of decomposing problems and incrementally solving them. Took help from AI if necessary.

Debugging: Have become quite skilled in debugging using the VS Code debugger as well as print statements.

Version Control: Have gained experience in using Git and GitHub to keep versions of my code.

Documentation: Have realized the importance of good documentation in the form of comments, README files and self-reflection.

GUIs: Have become quite competent with the various Java Swing components and their event handlers.

6. Understanding of Object-Oriented Programming concepts

Encapsulation

Have learned how to make use of access modifiers (private, public and protected) in order to control access to data and use getters/setters for data encapsulation.

Inheritance

Have gained skills in making effective use of inheritance to make class hierarchies and code reuse. Person → Student → Lecturer was a good example.

Polymorphism

Have become competent in the use of polymorphic references and method resolution during run-time in the Smart Home project.

Abstraction

Have learned how to abstract certain concepts and implement them in concrete classes. hiding implementation details from the user.

7. Future Learning Plans

From this experience, I will be able to:

1. Java Advanced Topics: Know about Collections Framework, Lambdas and Streams
2. Design Pattern: Know about various OOP Design Patterns and how to apply them
3. DB Integration: Be able to know about JDBC in Java DB Integration
4. Frame Work: Do some research about various Java Frameworks like Spring Boot
5. Mobile Application Development: Maybe learn about Android Mobile Application Development in Java
6. Contribution to Open-Source Project: Contribute to various Java Projects

8. Conclusion

These Object-Oriented Programming tutorials have helped me to be able to learn a lot about Java programming and Object-Oriented Programming. From being able to write basic programs to developing a full-fledged Java GUI Application.

The Quiz Battle game developed with the GUI is very interesting. Learning Java and OOP concepts has been very interesting indeed. This portfolio is proof of how much I have grown as a programmer.

This journey will help me become a good Java Programmer in the upcoming days.

9. GitHub Repository URL

Repository URL:

<https://github.com/pornadev08-ai/Dev-Porna-object-oriented-programming-tutorials01>

Submitted by: DEY PORNA RANI (202505010005)

Assignment 1 Rubric:

Section	Description	Weight (%)	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Excellent (5)
GitHub Repository Organization	Repository structure, folder organization, naming consistency, and use of a single repository.	4	Repository missing or multiple repositories used. Poor organization.	Repository exists but several required folders are missing or inconsistent.	Most required folders are available with acceptable organization.	Repository is organized with only minor issues.	Repository is well-organized, complete, consistently structured, and professionally maintained.
Tutorial Completeness	Inclusion of Week/Tutorial 1–10 source codes and supporting files.	6	Most tutorial work missing.	Less than half of the tutorials included.	Majority of tutorials included with some missing content.	Almost all tutorials included and functional.	All tutorials are complete, functional, and properly organized.
Repository	Quality, completeness	4	README	README	README	README is	README is

Documentation (README.md)	, clarity, and professionalism of the project documentation.		absent or contains almost no useful information.	contains limited information with poor formatting.	includes basic information but lacks clarity or completeness.	informative, organized, and mostly complete.	comprehensive, well-formatted, visually organized, and professionally written.
Self-Reflective Report	Reflection on learning experience, challenges, solutions, and inclusion of GitHub repository URL.	4	Reflection missing or extremely limited. Repository URL not included.	Reflection is brief with minimal discussion.	Reflection adequately discusses learning and challenges.	Reflection is clear, thoughtful, and supported with examples.	Reflection demonstrates deep critical thinking, meaningful learning, future improvement plans, and includes the repository URL.
Professionalism & Submission Compliance	Compliance with assignment requirements, report uploaded into	2	Multiple submission requirement	Several requirements missing or incomplete.	Most submission requirements fulfilled.	All requirements fulfilled with minor presentation	Fully complies with all requirements with excellent presentation

	repository, correct submission format, and overall presentation.		s not met.			ion issues.	n and professiona lism.
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